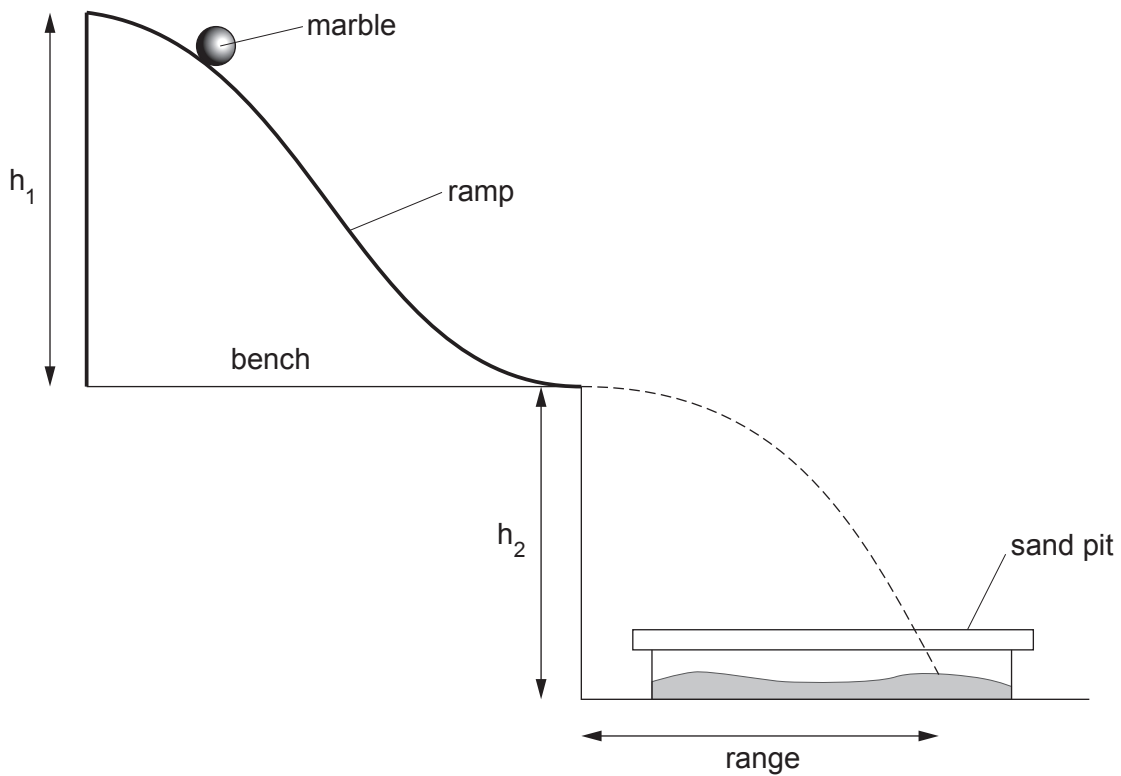


4. A model ski jump can be set up on a bench as shown in the diagram below.



The marble is released from rest at the top of a ramp. It leaves the bottom of the ramp with a certain velocity. It moves through the air and lands in a sand pit. The horizontal distance travelled in the air is called the **range**.

The velocity of the marble as it leaves the ramp is calculated using the equation:

$$\text{velocity} = \frac{\text{range}}{\text{time to travel the range}}$$

The acceleration of the marble down the ramp is calculated using the equation:

$$\text{acceleration} = \frac{\text{velocity of marble as it leaves the ramp}}{\text{time to travel down the ramp}}$$

- (a) The following measurements were recorded during one trial of the experiment.

Range = 45 cm

Time to travel the range = 0.5 s

Time to travel down the ramp = 1.2 s

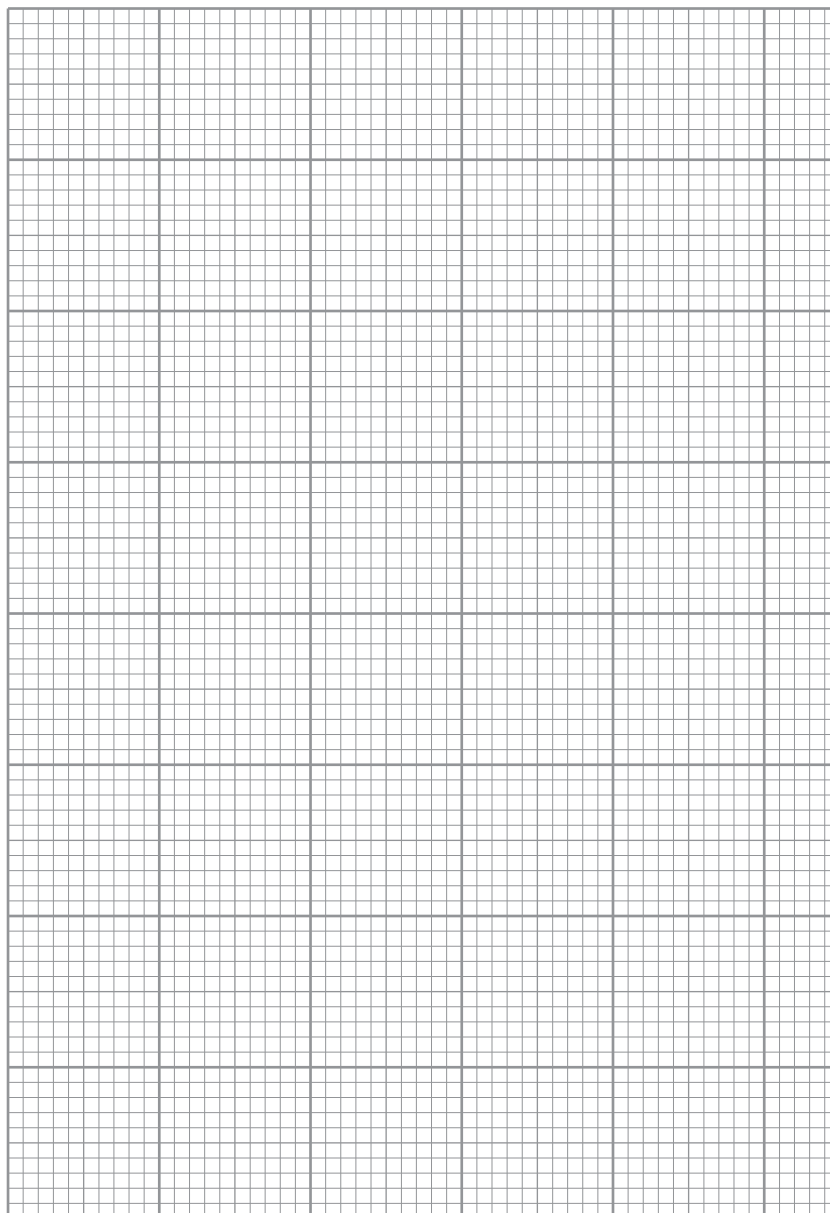
Use the equations and information above to calculate the acceleration of the marble down the ramp in m/s^2 . [4]

Acceleration = m/s^2

- (b) An experiment is then carried out to find out how the acceleration of the marble depends on the height h_1 . Throughout this experiment height h_2 is constant. The results of the experiment are shown in the table below.

Height h_1 (m)	Acceleration (m/s^2)
0	0
0.2	0.14
0.5	0.36
0.6	0.43
0.9	0.62
1.0	0.74

- (i) Use the data in the table to plot a graph on the grid below and draw a suitable line. [3]



- (ii) Use the graph to find the height required to produce an acceleration of 0.5 m/s^2 . [1]

Height =

- (iii) Describe how the experiment can be extended to determine how the range depends on the height h_2 . [2]

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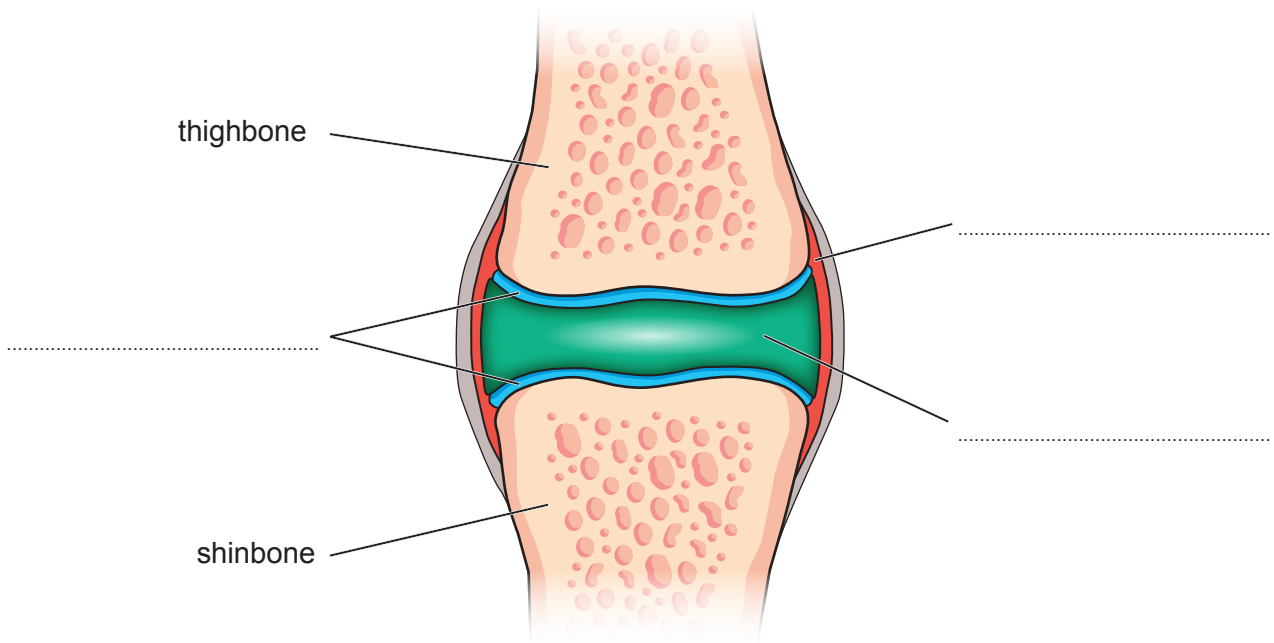
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- (c) Ski jumpers place their knee joints under considerable strain during their career. The knee is easily injured.

A simplified diagram of a knee joint is shown below.



- (i) **Complete** the labelling of the diagram above. [3]
- (ii) Explain why a ski jumper will not be able to compete if they damage the cartilage in their knee joints. [2]

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- (iii) An X-ray was taken of a ski jumper's leg following an accident.



State the name of this **type** of fracture.

[1]

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